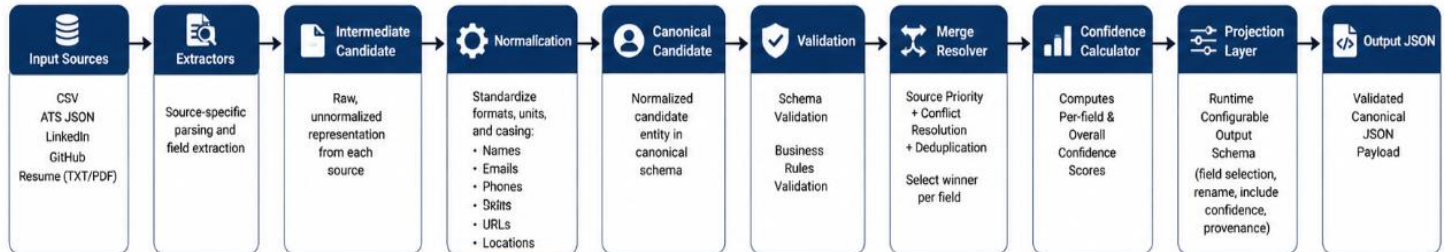


Multi-Source Candidate Data Transformer

(Technical design)

Pipeline overview:



Merges and normalizes candidate data from heterogeneous sources into a single canonical JSON profile with field-level confidence scores and provenance.

Canonical Output & Normalization Rules:

Schema: Candidate ID, Name, Emails, Phones, Location, Links, Headline, Experience, Education, Skills, Confidence, Provenance.

Normalization:

Field	Format
Phones	E.164
Dates	YYYY/YYYY-MM
Country	ISO-3166 Alpha-2
Skills	Canonical names
Emails	Lowercase
URLs	HTTPS

Merge & Conflict Resolution:

Priority: Recruiter CSV → ATS JSON → LinkedIn → GitHub → Resume PDF → Resume TXT

Rules: Higher priority wins • Deduplicate emails, phones & skills • Merge experience & education • Calculate confidence • Track provenance.

Confidence Score: Overall Confidence = $\sum(\text{Field Confidence} \times \text{Weight}) / \sum(\text{Weights})$

Runtime Configuration: The projection layer is fully configuration-driven. A runtime JSON file controls the output schema by selecting or renaming fields, enabling or disabling confidence and provenance, and defining the behaviour for missing values (null, omit, or error) without requiring any code changes

Scope:

Assumptions: Single candidate per run • Recruiter CSV preferred

Handled: Invalid contacts • Duplicates • Date normalization • Merge conflicts

Out of Scope: OCR • multi-candidate matching • ML confidence

